The Dynamics of Reference Point Formation: Evidence from a Dutch Pension Reform*

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Abstract

We study how individuals form reference points for retirement in the context of a Dutch pension reform that raised the Normal Retirement Age (NRA). Using full-population administrative data and a bunching approach, we document that a large retirement spike moves in lockstep with the legislated NRA, even though the NRA is not linked to any financial incentive to retire. Retirement behavior is sticky, however: significant residual bunching persists at the old NRA and responses to the new NRA build up gradually. We interpret these findings in a model of reference-dependent retirement. Calibrations suggest that one third of workers still hold the old NRA as their reference point in the short run, while two thirds immediately adopt the new NRA. After four years, virtually all individuals adjust their reference points to current policy. Our results imply that, despite gradual reference point adjustment, the large behavioral and fiscal impacts of increasing the NRA materialize relatively quickly.

Keywords: pension reform, retirement, bunching, reference-dependent preferences.
JEL codes: H55, J26, D91.

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1 Introduction

Population aging is putting unprecedented pressure on public pension systems across the developed world. The old-age dependency ratio in Europe and North America is projected to increase from around 33% today to more than 50% by 2050 (United Nations, 2024). At the same time, public pensions already constitute the largest item of public expenditure in most OECD countries (OECD, 2025). In response to these fiscal pressures, governments are enacting pension reforms aimed at encouraging later retirement. Understanding how retirement behavior responds to such reforms is crucial for assessing their fiscal and welfare effects.

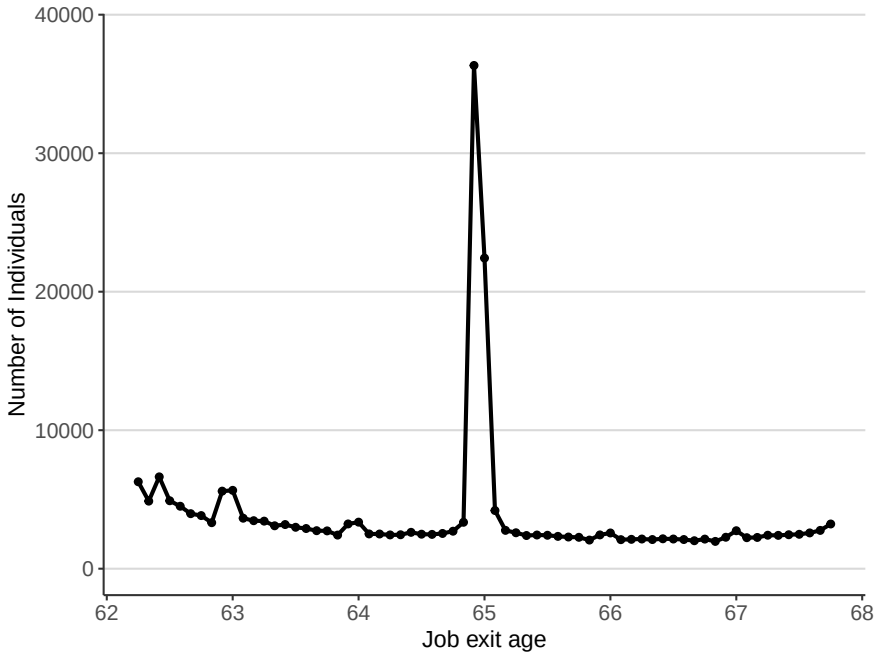
The most common type of pension reform increases statutory retirement ages, in particular the Normal Retirement Age (NRA) (Giupponi and Seibold, 2024). A large body of evidence shows that the NRA has a large impact on retirement behavior (e.g. Mastrobuoni, 2009; Seibold, 2021, 2024). Strikingly, these large responses occur even when the NRA is not linked to financial retirement incentives. Instead, the impact of the NRA has been attributed to its role as a *reference point*: the NRA is commonly communicated and perceived as the "normal" time to retire, and workers evaluate their retirement age relative to it (Behaghel and Blau, 2012; Seibold, 2021; Lalive et al., 2023; Reck and Seibold, 2026). Figure 1 illustrates this phenomenon in our setting. In the Netherlands, prior to recent reforms, the NRA had been set at age 65 for decades. The retirement age distribution exhibits an enormous spike precisely precisely in the month when individuals reach the NRA, even though there is no financial incentive to retire at this age.[1]

While the long-run effects of the NRA are well documented, less is known about the short- and medium-run dynamics of behavioral adjustment to reforms increasing the NRA. If the NRA affects retirement behavior because workers perceive it as a reference point, changing the NRA can be understood as government institutions framing retirement rules around a new reference point. In such a setting, the impact of a reform hinges on how individuals form their reference points and how quickly government policy can shift reference points. Do workers immediately adopt the new NRA set by the government as their reference point? Or do they still view the old NRA, which may have been in place for decades, as the "normal" time to retire?

In this paper, we study empirically how individual retirement behavior adjusts to an increase in the NRA, and how these behavioral responses are shaped by reference point formation around government policy. Our setting is a major pension reform in the Netherlands, which gradually raised the NRA from 65 to 67. The reform was implemented in a cohort-based way: starting with individuals born in 1948, successive birth cohorts face progressively higher NRAs, increasing in steps of one to four months. This creates a sequence of cohort groups who reach their new NRA at different points in time after the reform starts to take effect, allowing us to trace out the dynamics of behavioral adjustment. Importantly, the Dutch setting allows us to isolate the reference-point effect of the NRA, as neither the old nor the new NRA is linked to any financial incentive to retire. Our analysis is based on high-quality administrative data covering the universe of Dutch residents.

[1] In the Netherlands, public pension benefits do not depend on when an individual retires, and there is no earnings test for continued work after the NRA. See Section 2 for a detailed discussion of financial incentives around the NRA.

FIGURE 1: BUNCHING AT THE NORMAL RETIREMENT AGE



Notes: The figure shows the distribution of retirement ages in monthly bins among Dutch workers born between 1942 and 1947, who face a Normal Retirement Age (NRA) of 65. Retirement is defined as the last employment exit. See Section 2.2 for details on the data and sample.

We link various administrative datasets, allowing us to measure retirement behavior precisely at the monthly level for more than 1.3 million older workers.

Our empirical strategy relies on a bunching approach. For each cohort group, we estimate local retirement responses to the new NRA applying to these workers, as well as residual bunching at the old NRA of 65. This approach is well-suited to our purposes for several reasons. First, bunching at an age threshold that is not linked to any financial incentive provides a direct measure of its reference-point effect (Allen et al., 2017; Rees-Jones, 2018; Seibold, 2021). Second, measuring responses to both the old and the new NRA within the same group allows us to estimate the share of workers whose behavior has adjusted to the new NRA and the share who remain anchored at the old NRA. Third, the excess mass at each threshold maps directly into a model of reference-dependent retirement behavior, which allows us to further gauge the process of reference point formation.

Our empirical results reveals that the reform has a strong overall impact on retirement behavior, driven by a large retirement spike that moves in lockstep with the legislated NRA. We consistently find highly significant bunching precisely at the NRA in each cohort group, even though they face the NRA at different ages throughout the reform period.

However, two pieces of evidence suggest that retirement behavior is sticky in the short run. First, in the short-run, retirement responses are bifurcated: we observe bunching at the new NRA,

but also sizable residual bunching at the old NRA. Responses to the old NRA then decrease over time, but remain large and significant during the first three years after the reforms takes effect. Among cohort groups who reach their NRA four or more years after the reform, residual bunching at the old NRA dissipates. Second, bunching at the new NRA is smaller immediately after the reform than among the pre-reform cohorts, and it builds up gradually over time. These findings highlight the gradual nature of behavioral adjustment to the NRA increase. They also imply that individuals respond heterogeneously to the reform: while many workers adjust to the new NRA immediately, others continue to respond to the old NRA in the short run.

We interpret these findings through the lens of a model of reference-dependent retirement behavior, building on Seibold (2021) and Reck and Seibold (2026). In the model, the NRA serves as a reference point, generating bunching in the retirement age distribution. Informed by our empirical results, we model reference point formation as a process in which workers adopt either the old NRA or the new NRA as their reference point. Using our empirical bunching moments, the model then allows us to calibrate the dynamics of reference point adjustment over time. We find that reference points are sticky in the short run, with 36% of workers for whom the old NRA remains the reference point in the first year after the reform. However, the fraction of workers adopting the new legislated NRA as their reference point increases quickly over time, to 79% in the second and 93% in the third post-reform cohort group. After four years, the new NRA is virtually fully adopted as a reference point, with a stable share of more than 95%.

We also consider several alternative mechanisms and argue that none of them can account for the empirical patterns we document. First, as described above, financial retirement incentives cannot explain our findings. Second, prior literature has discussed both liquidity constraints and firm responses as potential contributors to retirement bunching at the NRA, but none of these can explain the sticky adjustment dynamics and residual bunching at the old NRA. Third, our findings are unlikely to be driven by a lack of information about the reform, as the NRA is highly salient in official communication, individuals are contacted by the social security administration before reaching their NRA, and survey evidence suggests that older workers are well-informed about their NRA.

Our results have two main policy implications. First, they speak to the speed of adjustment to pension reforms. While behavioral responses to an NRA increase are somewhat muted immediately after a reform, we find that they reach their full magnitude within four years. This implies that the positive fiscal effects of this type of pension reform materialize relatively quickly, which is important for policymakers seeking to stabilize public pension finances. Second, our findings show that government policy can influence reference points and norms around retirement. Although the NRA of 65 had been in place for decades, workers largely abandon within a few years after the government changes the age it designates as the NRA. This insight confirms that the framing of social security reform around salient age thresholds is a highly effective policy tool at the disposal of governments.

This paper contributes to three strands of literature. First, we contribute to the literature on

the impact of pension policies and reforms on retirement behavior (see Giupponi and Seibold, 2024, for an overview). A number of studies have empirically examined the large impact of statutory retirement ages such as the NRA on retirement behavior (e.g. Mastrobuoni, 2009; Staubli and Zweimüller, 2013; Cribb et al., 2016; Gruber et al., 2022; Lalive et al., 2023; Dolls and Krolage, 2023; Deshpande et al., 2024; Rabaté et al., 2024). The impact of pure financial retirement incentives has been found to be more modest, on the other hand (e.g. Brown, 2013; Manoli and Weber, 2016; Duggan et al., 2023). This discrepancy in the effectiveness of different policy instruments and excess retirements at the NRA have been attributed to the NRA serving as a reference point for retirement decisions (Behaghel and Blau, 2012; Seibold, 2021; Reck and Seibold, 2026).

Our main contribution to this literature is to study the dynamics of adjustment to pension reforms. While most existing studies focus on the overall or long-run effects of reforms, we trace out how behavioral responses evolve over time and across the retirement age distribution. Moreover, we relate these dynamics to the formation of reference points around statutory retirement ages, which is key for explaining the impact of NRA reforms. Closest to our paper, Deshpande et al. (2024) study the U.S. increase in the Full Retirement Age and find that employment exits exhibit stickiness at the old legislated age. We complement their findings by using high-frequency administrative data to quantify the speed of adjustment to pension reforms and by providing a conceptual rationale for stickiness. Another closely related paper is Rabaté et al. (2024), who study overall impact of the Dutch NRA reform on average retirement ages.

Second, we contribute to the literature on reference-dependent preferences (see O'Donoghue and Sprenger (2018) for a review). Field evidence documents reference-dependent behavior in a range of domains, including labor supply (Camerer et al., 1997; Farber, 2008; Crawford and Meng, 2011; Thakral and Tô, 2021), job search (DellaVigna et al., 2017), taxation (Homonoff, 2018; Rees-Jones, 2018), marathon running (Allen et al., 2017), housing demand (Genesove and Mayer, 2001; Andersen et al., 2022), and retirement (Seibold, 2021; Reck and Seibold, 2026). A key open and disputed issue in this literature is how individuals form their reference points. Broadly, two classes of reference point formation have been proposed. Forward-looking reference points are given by individuals' expectations about outcomes (Kőszegi and Rabin, 2006, 2007; Abeler et al., 2011; Crawford and Meng, 2011; Gill and Prowse, 2012; Bartling et al., 2015). Backward-looking reference points are determined by past outcomes or the status quo, and may adapt gradually to new circumstances (Baucells et al., 2011; DellaVigna et al., 2017; Thakral and Tô, 2021). In addition, reference points can be shaped by salient benchmarks, such as round numbers or institutional thresholds (Pope and Schweitzer, 2011; Allen et al., 2017; Seibold, 2021).

Our main contribution to this literature is to provide novel empirical evidence on reference point formation in a high-stakes field setting. While prior empirical work typically incorporates assumptions about reference point formation in structural models, we provide novel empirical evidence on the nature and timing of reference point adjustment in response to public policy. We find that reference points adjust gradually over time when a new salient benchmark is set by policymakers. Our results also imply that reference points are heterogeneous, with some workers

holding a reference point at the new NRA and others at the old NRA. Our results are consistent with a probabilistic model of reference point formation, but inconsistent with alternative models proposed in the literature in which individuals' reference points are a weighted average of past and present thresholds.

Third, we contribute to the methodological literature on bunching (see Kleven, 2016, for an overview), and in particular to work studying adjustment dynamics in response to shifting policy thresholds. This literature has largely focused on adjustment frictions in response to reforms of tax and transfer schedules, which cause individuals to adjust their behavior gradually when kinks are introduced, moved, or removed (Chetty et al., 2011; Kleven and Waseem, 2013; Gelber et al., 2020; Zaresani, 2020; Mavrokonstantis and Seibold, 2022). In these settings, stickiness arises within individuals, who do not immediately change their behavior relative to the past. Our contribution is to study adjustment dynamics in the context of retirement, where adjustment dynamics arise across individuals subject to successively different policies, and to show how dynamic bunching estimates can inform models of reference point formation.

The remainder of this paper is organized as follows. Section 2 describes the institutional background and our data. Section 3 presents our empirical analysis. Section 4 interprets our findings in a model of reference-dependent retirement behavior and presents calibration results. Section 5 discusses alternative mechanisms, and Section 6 concludes.

2 Institutional Background and Data

2.1 Institutional Background

The Dutch pension system rests on three pillars. The first pillar consists of public pay-as-you-go pensions (*AOW*), financed by social insurance contributions. Benefit entitlements depend only on the length of residence in the Netherlands: each year of residence accrues two percent of the full benefit, which is paid to individuals with at least 50 years of residence. The full benefit is tied to the statutory minimum wage.[2] First-pillar pensions are paid automatically from the Normal Retirement Age (NRA, or *AOW age* in Dutch), and they can neither be claimed early nor deferred. There is no earnings test, so that individuals can freely combine pension receipt with continued employment. Importantly for our purposes, reaching the NRA therefore does not create a financial incentive to stop working.[3] It is worth noting that many employment contracts end automatically at the NRA, such that firms and workers have to actively renew the contract in order to continue employment. After having been set at 65 for decades, the NRA has been gradually increased to 67 in recent years (see below).

[2] A single pensioner receives 70% of the net minimum wage, while couples jointly receive 100%. The monthly minimum is set by the government and updated twice per year. As of July 2026, the monthly minimum wage is EUR 2338.

[3] Individuals above the NRA no longer pay first-pillar social insurance contributions, which, if anything, increases the financial return to working beyond the NRA.

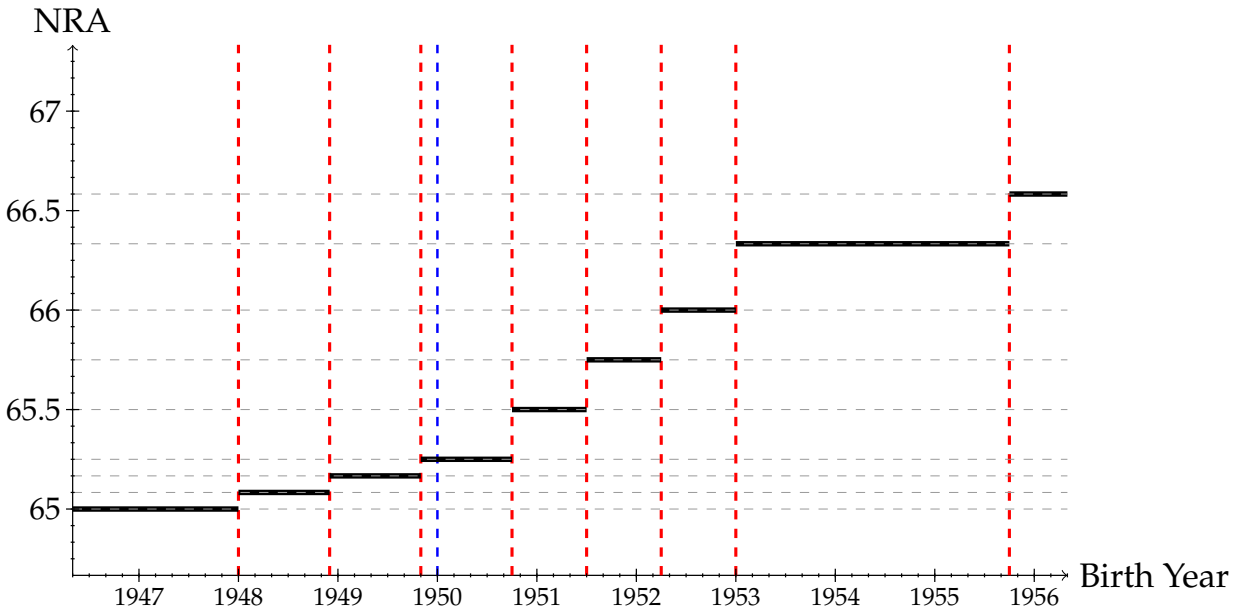
The second pillar consists of employer-based, funded pension schemes. Participation is mandatory for employees, and most plans are organized at the sector level, with some large firms operating their own funds. Benefits depend on contributions made by workers and employers. Second-pillar pensions can be claimed from an Early Retirement Age (ERA), which differs across sectors, and for most workers they constitute the only source of pension income before the NRA. They are a quantitatively important component of retirement income: the median retiree's replacement rate from second-pillar pensions is 29%, compared to 39% from first-pillar pensions (Knoef et al., 2016). The third pillar consists of voluntary private retirement savings, which are subsidized through the tax system but play a comparatively minor role, accounting for around 5% of total pension income.

The NRA as an Institutional Reference Point. Like in many other countries, the NRA occupies a central place in how retirement rules are communicated by institutions and discussed in the Netherlands. Government information material is organized around the NRA: official websites of the central government and the social security administration prominently display the NRA and offer online tools where individuals can enter their birth date to obtain their NRA. Each individual's NRA is also displayed in personal online accounts where pension information is shown, and individuals receive a letter in the year before they reach the NRA to initiate public pension payments. Accordingly, the NRA has been one of the most salient and contested issues in public debates around retirement policy. Survey evidence suggests that Dutch workers view the NRA as a "normal" time to retire that shapes their retirement plans, with many expecting to retire exactly at this age (Alessie et al., 2011; de Grip et al., 2013; Vermeer et al., 2019). In line with the NRA serving as a reference age for retirement, a large share of workers are observed to retire precisely at this age. Figure 1 illustrates this large retirement concentration for cohorts 1942 to 1947, for whom the NRA was 65.

The 2011 Reform. In response to concerns about the long-run fiscal sustainability of the public pension system, the Dutch government enacted a reform in 2011 that raised the NRA from 65 to 67.[4] The increase was phased in gradually from 2013 onward and is cohort-based, such that individuals born in 1948 and later are affected. Figure 2 illustrates the reform schedule. The NRA increases by one month for each of the first three steps at birth date cutoffs January 1948, December 1948, and November 1949. The pace of the reform then accelerates, with increases to 65 and 6 months, 65 and 9 months, 66 and 66 and 4 months at birth date cutoffs October 1950, July 1951, April 1952 and January 1953. After remaining at 66 and 4 months until birth cohort August 1955, the NRA finally increases to 66 and 7 months, 66 and 10 months and 67 at birth date cutoffs September 1955, June 1956 and March 1957. From 2028 onward, the NRA will be linked to life expectancy, with the first formulaic increase to 67 and 3 months affecting cohorts 1961 and younger. Importantly, the 2011 reform did not change benefit levels or modify financial retirement incentive around the new NRA.

[4]The reform was passed in 2011. Modifications in 2015 and 2019 first accelerated and then slowed down the NRA increase.

FIGURE 2: THE 2011 REFORM – NORMAL RETIREMENT AGE BY BIRTH DATE



Notes: The figure shows the Normal Retirement Age by birth date under the 2011 pension reform. The red dashed vertical lines demarcate the birth date cutoffs at which the NRA increases. The blue dashed vertical line demarcates the birth date cutoff of the 2006 reform, which reduced financial incentives for early retirement.)

Other Reforms. A second reform enacted in 2006 changed the rules of employer-based pensions. Its main aim was to curb early retirement, which had largely been facilitated by second-pillar provisions. The reform encompassed a bundle of measures in order to reduce financial incentives for early retirement, most importantly cutting early retirement benefits and introducing actuarially fair adjustment of second-pillar pensions. It applied only to workers born in January 1950 or later, and it substantially increased the average retirement age of affected cohorts (Lindeboom and Montizaan, 2020; Rabaté et al., 2024; Oral et al., 2026). While the cohort cutoff of the 2006 reform does not coincide with any of the NRA cutoffs we study in this paper, potential interaction effects between the two reforms are relevant for interpreting our empirical results. The 2006 reform generally induces later retirement and has been shown to increase the share of workers retiring around the NRA (Oral et al., 2026). We discuss this interaction in detail in Section 3.2.

Further policy changes that took place during our sample period include changes to private retirement savings subsidies in 2006 and 2015, changes to tax credits for older workers in 2012 and 2018, and the removal of a pension supplement for non-working spouses in 2015. As these changes are not tied to the cohort steps of the NRA increase, they are unlikely to affect our analysis.

2.2 Data

We use administrative data on the universe of Dutch residents provided by Statistics Netherlands (CBS). We construct our dataset by linking several administrative registers via pseudonymized personal identifiers, including civil registration data, employer-employee matched labor market data, and income and wealth tax data. Our data contains monthly information on individual labor market states and income sources, as well as sociodemographic characteristics such as date of birth, gender, marital status, and country of birth. Our analysis period is 1999 to 2022.

Our main outcome is a worker's retirement age, which we define as the age of last employment exit. We focus on cohorts born between 1942 and 1954, for whom we can observe retirement ages between 57 and 67. Cohorts 1942 to 1947 face an NRA of 65 and serve as the pre-reform baseline, while the cohorts born between 1948 and 1954 are affected by the gradual NRA increases due to the 2011 reform. We further restrict the sample to individuals who are employed for at least one month at or after age 57, such that their last employment exit can be interpreted as a retirement event. Our final sample comprises 1,778,777 individuals.

Table 1 presents summary statistics of our pooled sample of cohorts born between January 1942 and December 1954. 38% of individuals in our sample are female, reflecting relatively low labor force participation among women in these cohorts. 77% are married or in a civil partnership, and 86% are Dutch-born. Average annual labor income at age 57 is EUR 33,752, with a median of EUR 29,293. The average retirement age in our sample is 63.5 years. 14% of individuals retire exactly at their NRA, 62% retire before the NRA, and the remaining 24% retire after the NRA. In Appendix Table A1, we additionally show summary statistics separately for each cohort facing different a different NRA.

3 Empirical Analysis

3.1 Descriptive Evidence

The 2011 reform gradually increases the NRA around a sequence of birth date cutoffs. We begin by documenting how retirement behavior evolves across the cohorts affected by the reform. Panel (a) of Figure 3 plots the average retirement age by month of birth. The vertical lines demarcate the birth-date cutoffs at which the NRA increases. Among pre-reform cohorts born between 1942 and 1947, who face an NRA of 65, the average retirement age is relatively stable at around 63 years. Starting with the first cohorts affected by the reform, the average retirement age increases markedly, reaching around 63.5 years among individuals born in 1949. A particularly large increase in the average retirement occurs around the January 1950 cutoff of the 2006, which increased financial incentives for later retirement (Oral et al., 2026). Among cohorts 1950 to 1953, we again observe a gradual increase in the average retirement age, in line with the increasing NRA. This strong increase in the average retirement age among cohorts subject to the NRA increase provides a first indication of the impact of the reform. However, the discontinuous increase around the Jan-

TABLE 1: SUMMARY STATISTICS

	(1) Mean	(2) S.D.	(3) Median
Female	0.38	0.48	0.00
Married or civil partnership	0.77	0.42	1.00
Dutch-born	0.86	0.34	1.00
Annual labor income at 57	33,751.75	40,963.95	29,293.00
Year of birth	1949.02	3.62	1949.17
Retirement age	63.54	3.24	63.83
Normal Retirement Age (NRA)	65.42	0.52	65.17
Retirement at the NRA	0.17	0.37	0.00
Retirement before the NRA	0.59	0.49	1.00
Retirement after the NRA	0.24	0.43	0.00
Number of observations	1,778,777		

Notes: The table shows summary statistics of the pooled sample of all cohorts born between January 1942 and December 1954. Retirement age is the age of last employment exit. "S.D." denotes standard deviation.

uary 1950 also points at the 2006 reform, whose interaction with the NRA increase we have to take into account.

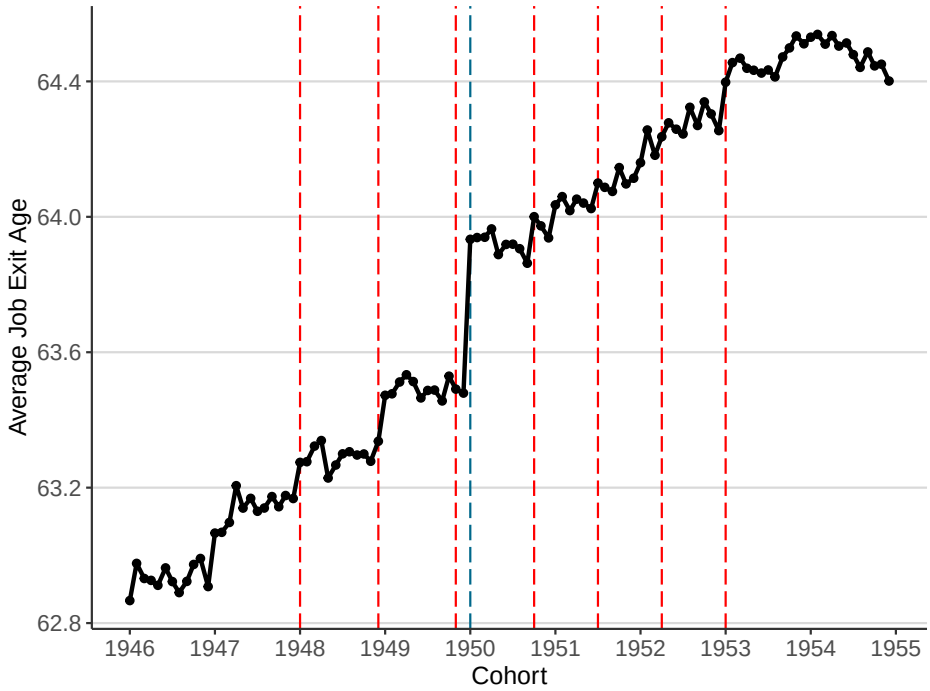
In order to understand the anatomy of these changes in retirement behavior during our sample period, Panels (b) and (c) of the figure show how the distribution of retirement ages changes throughout the cohort window from Panel (a). In Panel (b), we observe a large spike in retirements precisely at the NRA of 65 among pre-reform cohorts 1942 to 1947. In Panel (c), the large retirement spike has moved precisely to the new NRA of 66 and 4 months among the latest post-reform cohorts 1953 to 1955. This illustrates that the gradual increase in the average retirement from Panel (a) is largely driven by the retirement spike at the NRA moving along with the legislated NRA. In addition, there is a clear reduction in early retirements due to the 2006 reform. Our main empirical analysis below isolates the impact of the 2011 reform on the retirement age distribution, and trace out the dynamics of behavioral responses to the NRA increases.

3.2 Bunching Estimation

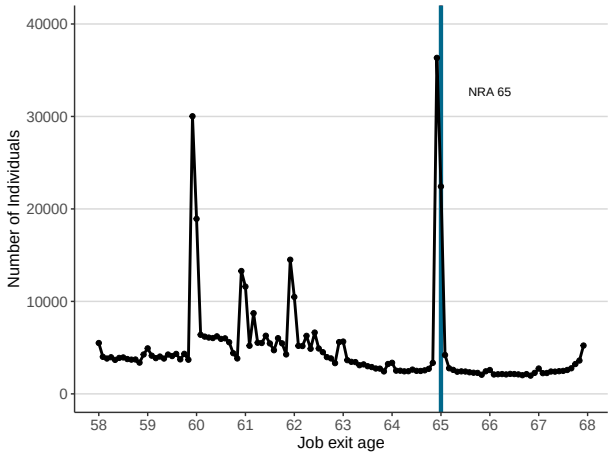
Method. In order to quantify retirement responses to the old and new NRA, we use a bunching approach (Saez, 2010; Chetty et al., 2011; Kleven, 2016). The goal is to measure the excess mass at an age threshold R^*, relative to a counterfactual distribution that would be observed in the absence of any policy change linked to the threshold. We denote the bunching mass, that is, the observed local spike in the density at R^* by B. We scale the bunching mass by the counterfactual density at

FIGURE 3: RETIREMENT BEHAVIOR ACROSS COHORTS

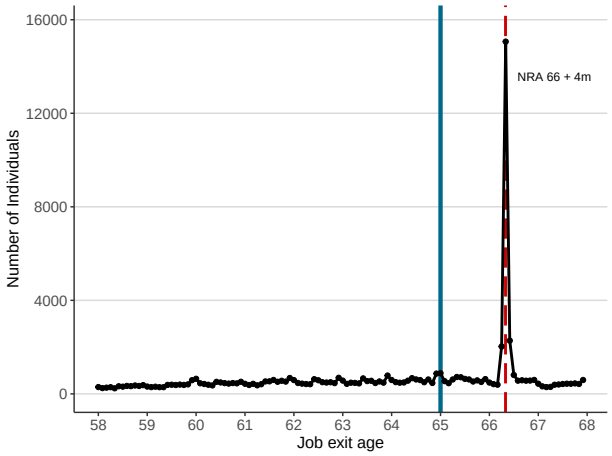
(a) Average Retirement Age by Birth Date



(b) Retirement Age Distribution: Pre-Reform Cohorts (NRA 65)



(c) Retirement Age Distribution: Post-Reform Cohort Group (NRA 66 + 4 months)



Notes: Panel (a) shows individuals' average retirement age by their month of birth. The red vertical lines demarcate the birth-date cutoffs at which the NRA increases under the 2011 reform (see Figure 2). The blue vertical line demarcates the the birth-date cutoff of the 2006 reform, which changed financial retirement incentives. Panels (b) and (c) show the distribution of retirement ages in monthly bins among individuals born between 1942 and 1947, who face an NRA of 65, and among individuals born in 1953 and 1954, who face an NRA of 66 and 4 months, respectively. The blue and red vertical lines demarcate the old NRA of 65 and the new NRA, respectively.

the threshold, $h(R^*)$, to obtain the excess mass $b = B/h(R^*)$. This relative measure is comparable across cohort groups of different sizes. Moreover, it has an intuitive interpretation: b approximates the retirement age response of the marginal bunching individual, that is, the individual who would have retired furthest away from the threshold in the counterfactual (Seibold, 2021).

We estimate the counterfactual distribution following Chetty et al. (2011) and Seibold (2021). Specifically, we fit a flexible polynomial to the observed retirement age distribution, excluding a bunching region around the threshold and controlling for other features of the distribution:

$$c_j = \sum_{i=0}^{p} \beta_i \left(R_j\right)^i + \sum_{i=R_L}^{R_U} \gamma_i \mathbb{1}\left[R_j = i\right] + \sum_{r \in \mathcal{Q}} \rho_r \mathbb{1}\left[\frac{R_j}{r} \in \mathbb{N}\right] + \sum_{k \in K} \theta_k \mathbb{1}\left[R_j = k \wedge R_j \notin \left[R_L, R_U\right]\right] + \nu_j \tag{1}$$

where c_j is the number of individuals retiring in monthly age bin j and R_j is the corresponding retirement age, measured in months. The first term is a polynomial of degree p in retirement age; we use $p = 9$ in our baseline specification. The second term includes a dummy for each bin within the bunching region $[R_L, R_U]$ around the threshold. The third term controls for round-number effects. In our context, these are excess retirements at round ages, that is, at full years of age. Since we measure retirement ages in months, this corresponds to $\mathcal{Q} = \{12\}$. The last term includes dummies for other age thresholds $k \in K$ outside the bunching region, such as the NRA in post-reform cohort groups when estimating bunching at the old NRA, and vice versa. The counterfactual distribution is obtained as the predicted values from equation (1), omitting the contribution of the dummies for the bunching region:

$$\hat{c}_j = \sum_{i=0}^{p} \hat{\beta}_i \left(R_j\right)^i + \sum_{r \in \mathcal{Q}} \hat{\rho}_r \mathbb{1}\left[\frac{R_j}{r} \in \mathbb{N}\right] + \sum_{k \in K} \hat{\theta}_k \mathbb{1}\left[R_j = k \wedge R_j \notin \left[R_L, R_U\right]\right] \tag{2}$$

The bunching mass is then given by the difference between the observed and the counterfactual number of retirements within the bunching region:

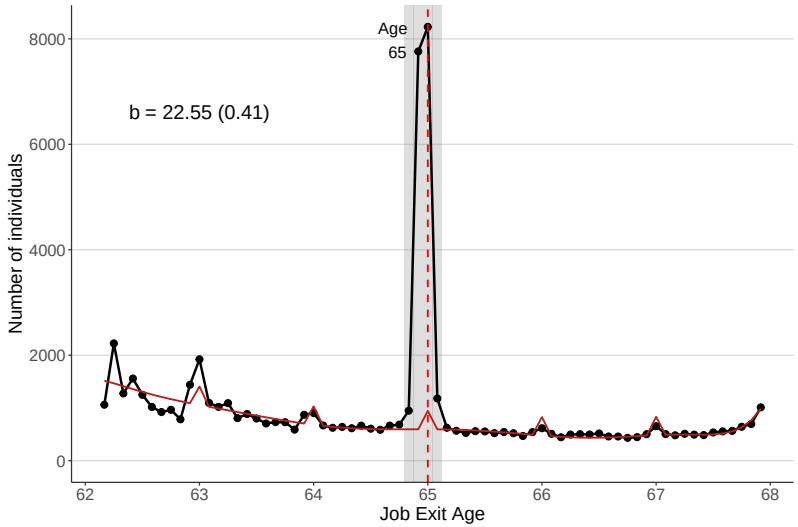
$$\hat{B} = \sum_{j:\, R_j \in [R_L, R_U]} \left(c_j - \hat{c}_j\right) \tag{3}$$

and the excess mass is $\hat{b} = \hat{B}/\hat{c}_0$, where $\hat{c}_0$ denotes the average counterfactual density within the bunching region. Standard errors are computed by bootstrapping.

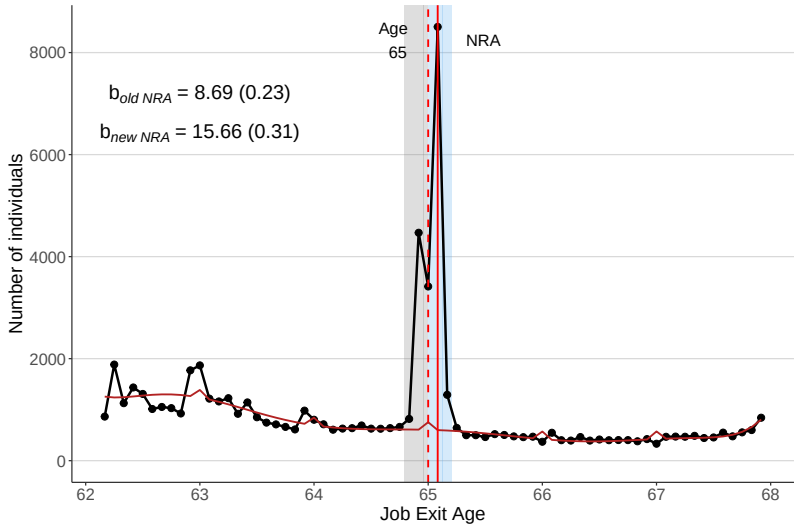
For each post-reform cohort group t, we estimate two excess masses: bunching at the new NRA, b_t^t, and residual bunching at the old NRA of 65, b_0^t. The superscript indexes the cohort group and the subscript indexes the threshold at which bunching is measured. The key identification assumption underlying the bunching approach is that the retirement age distribution would be smooth around the threshold in the absence of the NRA (except potential round-number effects, which we control for). This assumption is supported by the observation that the retirement age distribution is smooth around the new NRA among pre-reform cohorts, and around the old NRA among later post-reform cohorts, once short-run residual bunching dissipates.

FIGURE 4: DYNAMICS OF BUNCHING AT THE OLD AND NEW NRA

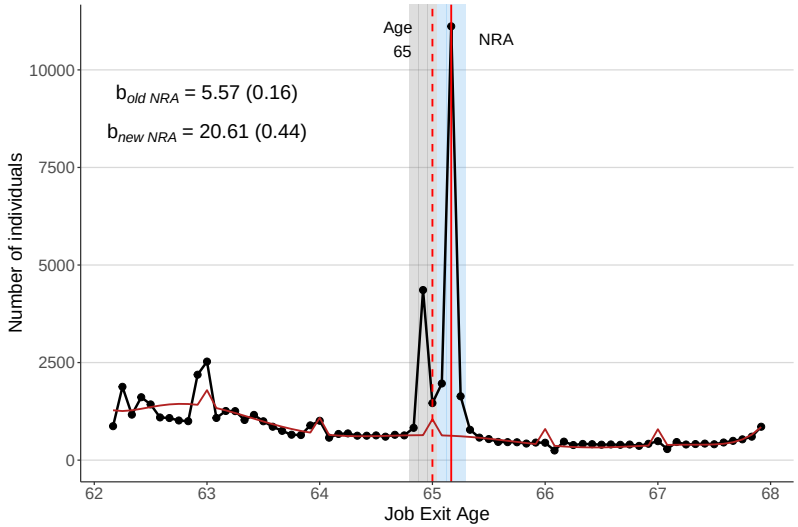
(a) $t = 0$: NRA 65 (born Jan. 1947 – Dec. 1947)



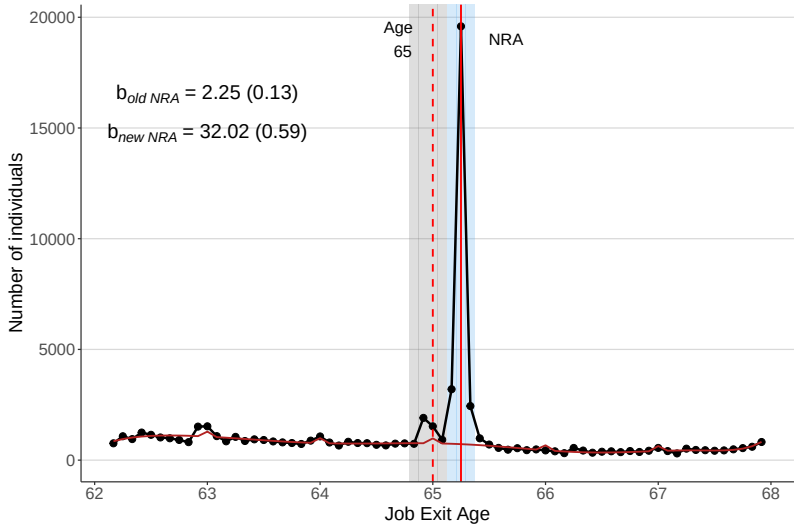
(b) $t = 1$: NRA 65 + 1 month (born Jan. 1948 – Nov. 1948)



(c) $t = 2$: NRA 65 + 2 months (born Dec. 1948 – Oct. 1949)



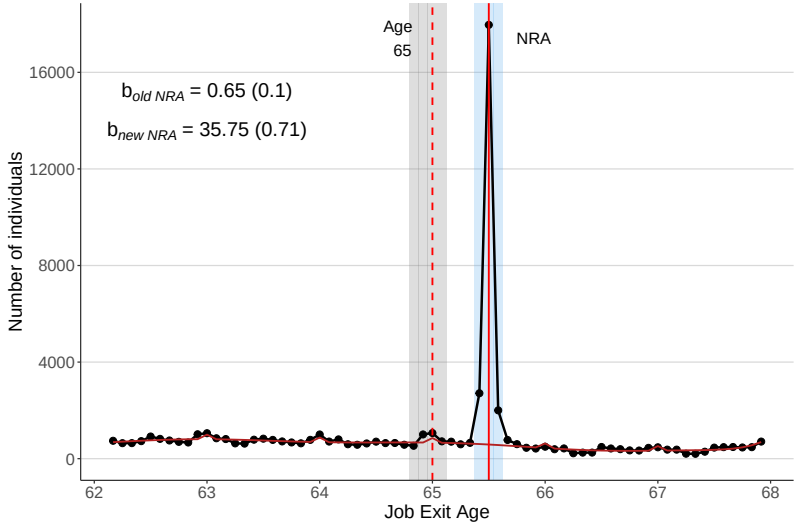
(d) $t = 3$: NRA 65 + 3 months (born Nov. 1949 – Sep. 1950)



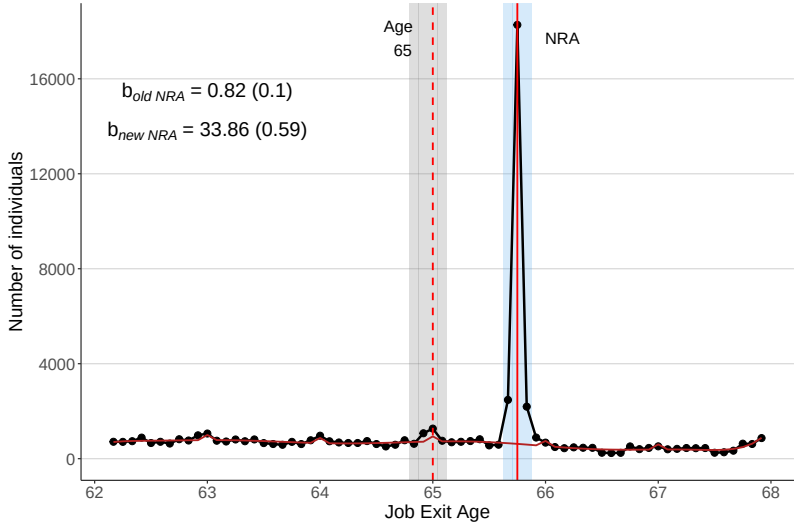
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FIGURE 4: DYNAMICS OF BUNCHING AT THE OLD AND NEW NRA (CONTINUED)

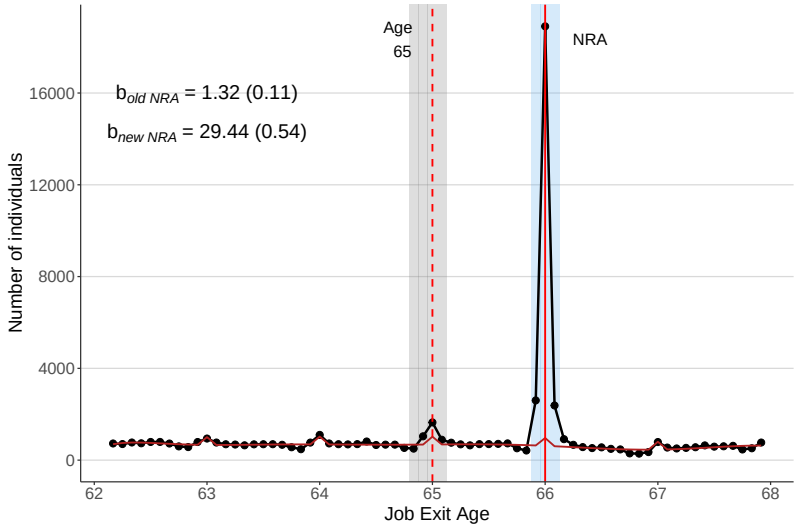
(e) $t = 4$: NRA 65 + 6 months (born Oct. 1950 – Jun. 1951)



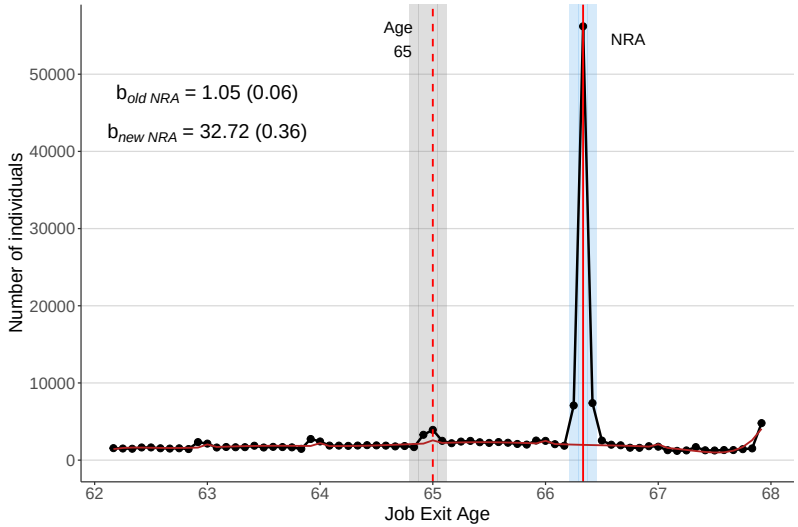
(f) $t = 5$: NRA 65 + 9 months (born Jul. 1951 – Mar. 1952)



(g) $t = 6$: NRA 66 (born Apr. 1952 – Dec. 1952)



(h) $t = 7$: NRA 66 + 4 months (born Jan. 1953 – Aug. 1955)



Notes: The figure shows the distribution of retirement ages in monthly bins for each NRA cohort group, as indicated by the panel titles. Retirement is defined as the last employment exit at age 57 or older. In each panel, the black line depicts the observed distribution and the red line depicts the estimated counterfactual distribution, obtained by fitting a polynomial of degree 9 to the observed distribution as described in Section 3.2. The text boxes display the estimated excess mass at the new NRA and, for post-reform cohort groups, at the old NRA of 65, with standard errors in parentheses. Note that the January 1950 cohort cutoff of the 2006 reform falls within cohort group $t = 3$.

Results. Figure 4 displays the observed retirement age distributions together with the estimated counterfactual distributions for each cohort group. Panel (a) shows the pre-reform cohort born in 1947 whose NRA is 65. We observe a strong concentration of retirements precisely at the NRA.[5] The estimated excess mass of 22.6 is large and highly significant, indicating a propensity to retire at the NRA more than 22 times larger than what the counterfactual density would predict. This magnitude of excess retirements is well in line with prior estimates of retirement bunching at the NRA in other settings (Seibold, 2021).

Panels (b) to (h) turn to the post-reform cohorts. In Panel (b), the first post-reform cohort group (January to November 1948), facing an NRA of 65 and 1 month, exhibits substantial bunching at this new NRA. The retirement pattern around the NRA is bifurcated, though, with large spikes both at the new and the old NRA. The estimated excess mass at the new NRA is 15.7, and that at the old NRA is 8.7. Panel (c) displays the retirement age distribution among the second post-reform cohort group (December 1948 to October 1949), whose NRA is 65 and 2 months. Again, we observe large bunching at the new NRA (excess mass 20.6), but also sizable residual bunching at the initial NRA of 65 (excess mass 5.6). In Panel (d), workers born between November 1949 and September 1950 also exhibit large bunching at their NRA of 65 and 3 months (excess mass 32.0), while there is a smaller but still clearly visible spike at the old NRA of 65 (excess mass 2.3). In Panels (e) to (h) of the figure, a large retirement spike continues to move precisely in lockstep with the legislated NRA of 65 and 6 months, 65 and 9 months, 66, and 66 and 4 months among workers born between October 1950 and August 1955. Only a small spike remains at the old NRA among these younger cohorts, which is not clearly distinguishable from round-number effects.

Figure 5 summarizes our excess mass estimates across all cohort groups. Panel (a) shows bunching at the new NRA, and Panel (b) shows residual bunching at the old NRA. The figure highlights three main empirical findings from the bunching estimation. First, a large retirement spike tracks precisely the legislated NRA throughout the reform period. Panel (a) shows that, even though different cohorts face the NRA at different ages, we estimate large excess mass at the NRA consistently across all cohorts in our data. Second, in terms of magnitude, the excess mass is smaller immediately after the reform than among pre-reform cohorts, but then becomes gradually larger among later cohorts. Particularly large bunching emerges among cohorts t =3 to 7, but this has to interpreted with some caution, as the 2006 reform reduced incentives for early retirement and led to more retirements at the NRA among workers born in 1950 and later (Rabaté et al., 2024; Oral et al., 2026).

The third key empirical result illustrated in Panel (b) is that there is substantial stickiness of retirements at the old NRA in the short-run. In the first year after the 2011 reform becomes effective, there is large residual bunching at the old NRA, with an excess mass of a magnitude comparable to bunching at the new NRA. Residual bunching at the old NRA decreases over time, with large

[5]Note that among the earlier cohorts in our data, bunching at the NRA is spread across two monthly age bins, 64 and 11 months and 65. This occurs because until 2012, public pension payments start on the first day of the month when the individual reaches the NRA, such that individuals who retire precisely at the NRA often exit employment on the last day of the previous month. From 2012 onward, public pension payments start on the exact day the individual reaches the NRA, such that employment exit bunching occurs within the month of the NRA.

and significant excess mass estimates up until workers born in 1950 (cohorts $t = 1$ to 3), who reach the NRA in the first three years after the reform becomes effective. Among those cohorts facing an NRA of 65 and 6 months or higher, who reach the NRA four or more years after the reform, there is no longer any clear residual bunching at the old NRA.

Overall, these empirical results highlight the gradual nature of behavioral adjustments to the NRA increase and the stickiness of retirement choices in the short-run. Our findings also imply that dynamic responses to the NRA are heterogeneous, with some workers adjusting immediately to the NRA, while others keep retiring at the old NRA. In the next section, we interpret these dynamics through the lens of a model of reference-dependent retirement behavior.

4 Model and Parameter Estimation

Our empirical analysis documents three key patterns. First, the large retirement spike at the NRA moves along with the legislated NRA. Second, bunching at the new NRA builds up gradually over the first cohorts affected by the reform. Third, a sizable residual spike remains at the old NRA of 65 in the short run, which then largely disappears after around three years. In this section, we argue that these patterns are well explained by reference-dependent behavior, and that the observed dynamics are informative about how individuals adopt reference points for retirement. We formalize this mechanism in a simple model of reference-dependent retirement behavior with gradual reference point adjustment. We then use the model to calibrate the speed at which workers adopt an NRA that is shifted by public policy as their reference point. In Section 5, we discuss some potential alternative mechanisms behind our empirical findings.

4.1 The NRA as a Reference Point for Retirement

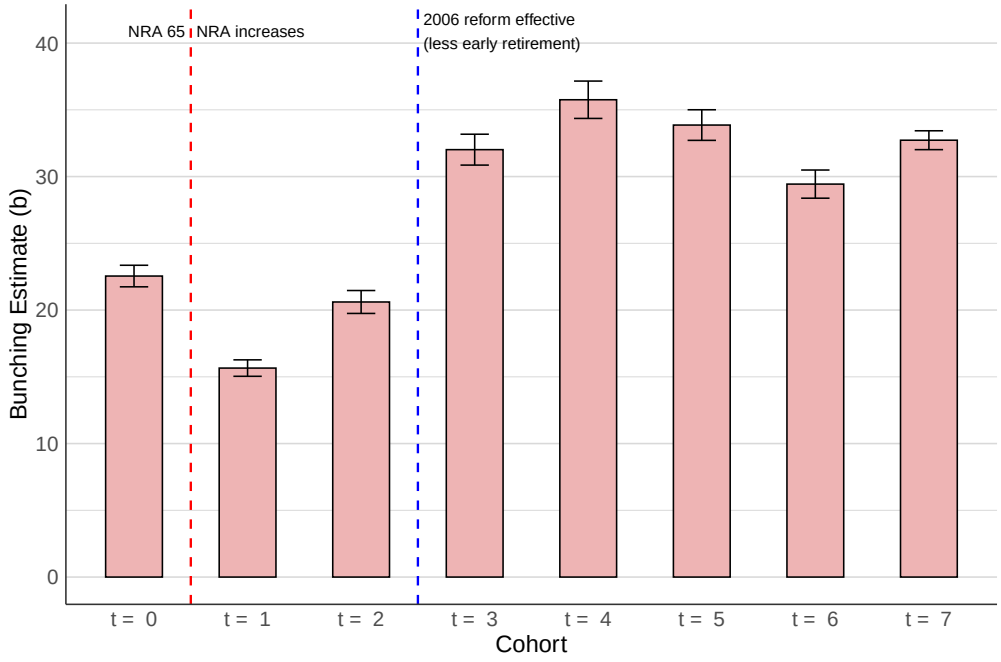
A large body of evidence suggests that the NRA serves as a reference point for retirement decisions. Excess retirements at the NRA are a long-standing empirical regularity across countries (Lumsdaine et al., 1996). Recent empirical literature documents that bunching at the NRA cannot be explained by financial incentives, and instead is consistent with reference-dependent preferences, under which workers evaluate their retirement age relative to the NRA as a "normal" time to retire (Behaghel and Blau, 2012; Seibold, 2021; Gruber et al., 2022; Lalive et al., 2023; Dolls and Krolage, 2023; Reck and Seibold, 2026). Survey evidence further indicates that workers perceive the NRA as reference point, including in the Dutch context (Vermeer et al., 2019).[6]

Reference dependence also offers a natural explanation for the dynamics we document. We find that some workers bunch at the new NRA, while others bunch at the old NRA, both of which are not linked to financial retirement incentives. In our context, the old NRA had been in place for several decades and was deeply ingrained as the "normal" age of retirement (see Section 2.1).

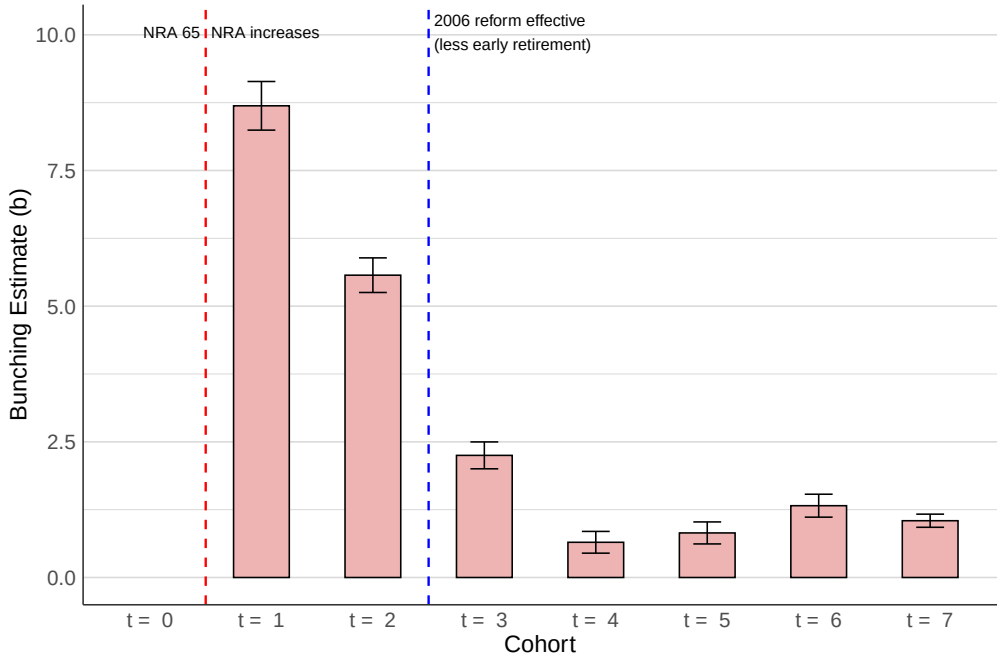
[6]Prior literature also considers a range of alternative explanations for the concentration of retirements around statutory retirement ages, including liquidity constraints and firm responses, but finds less evidence in favor of those (see Section 5).

FIGURE 5: BUNCHING ESTIMATES

(a) Bunching at the New NRA



(b) Residual Bunching at the Old NRA



Notes: The figure shows estimated excess masses across NRA cohort groups. Panel (a) shows bunching at the new NRA and Panel (b) shows residual bunching at the old NRA of 65. Each point corresponds to an excess mass estimate with its 95% confidence interval, based on bootstrapped standard errors. Retirement is defined as the last employment exit at age 57 or older.

Evidence from other domains suggests that reference points adapt gradually to new circumstances (DellaVigna et al., 2017; Thakral and Tô, 2021). If retirement reference points are shaped by past government communication and public debates, they need not adjust immediately when the NRA changes. Our empirical results suggest that some workers adopt the new NRA as a reference point, while others continue to treat the old NRA at 65 as their reference point, at least in the short run.

4.2 Model

Setup. We adopt a simple static model of reference-dependent retirement behavior, which is closely in line with prior applied work and well-suited to parsimoniously capture bunching at the reference point (Seibold, 2021; Reck and Seibold, 2026). Individuals choose their retirement age, where later retirement causes disutility from working longer, and they additionally evaluate their retirement age relative to a reference point. Preferences are given by[7]

$$U = u(C) - v(R, n) - \mathbb{1}\left(R \geq \hat{R}\right) \cdot \widetilde{\lambda}\left(R - \hat{R}\right) \tag{4}$$

where C is lifetime consumption and R is the worker's retirement age relative to a career starting age normalized to 0. $u(C)$ denotes utility from consumption and $v(R, n)$ is disutility from work. The parameter n reflects earnings ability at old age, where low ability increases disutility from postponing retirement. The parameter $\widetilde{\lambda} \geq 0$ captures the strength of reference dependence.

In line with prior work, we model reference dependence as loss aversion over lifetime leisure.[8] Intuitively, marginal disutility from increasing labor supply beyond the retirement reference point $\hat{R}$ is greater than marginal disutility from approaching $\hat{R}$ from the left, and the parameter $\widetilde{\lambda}$ determines the size of this kink in the utility function.

Individuals face a lifetime budget constraint

$$C = w(1 - \tau)R \tag{5}$$

where w denotes the gross wage per period and τ the implicit tax rate on continued work, capturing taxes, social insurance contributions, and the change in pension wealth from working longer. Earnings ability n is heterogeneous across individuals and distributed according to a smooth density $f(n)$. In the absence of reference dependence ($\lambda = 0$), this generates a smooth counterfactual distribution of retirement ages $h_0(R)$.

Bunching at the Reference Point. Reference dependence introduces a kink in individuals' preferences at $\hat{R}$: above the reference point, the marginal return to working longer is reduced by λ.

[7] The simple model we consider here can be interpreted as a reduced form of a dynamic model of labor supply. The static version is sufficient to explain key empirical retirement patterns (see Burtless, 1986; Manoli and Weber, 2016) and provides a convenient way to model reference dependence in retirement behavior. The main implicit assumption behind the static model is that all choice-relevant information is revealed before the initial period (e.g. age 55). See Reck and Seibold (2026) for a more detailed discussion.

[8] Out of a range of alternative formulations of reference dependence, loss aversion over lifetime leisure provides the best fit to the data in the retirement context. See Seibold (2021) and Reck and Seibold (2026) for a detailed discussion.

Analogous to a kink in the budget set (Saez, 2010; Kleven, 2016), this induces individuals to bunch at the reference point. As Seibold (2021) shows, one can identify a marginal bunching individual, whose indifference curve would be tangent to the budget line at some retirement age R^* without reference dependence, and who is tangent exactly at $\hat{R}$ with reference dependence. Thus, the bunching mass B at the reference point is given by

$$B = \int_{\hat{R}}^{R^*} h_0(R)dR \approx h_0(\hat{R})(R^* - \hat{R})$$

where $h_0(\hat{R})$ is the height of the counterfactual retirement density at $\hat{R}$. In other words, all individuals who would retire in an interval $[\hat{R}, R^*]$ in the absence of reference dependence now retire exactly at $\hat{R}$.

In order to obtain a closed-form solution, we can adopt the standard parametrization of the bunching literature, with quasi-linear utility $u(C) = C$ and an isoelastic disutility of work $v(R, n) = \frac{n}{1+1/\varepsilon}\left(\frac{R}{n}\right)^{1+1/\varepsilon}$. Individuals' optimal retirement age is then given by $R = n\left[w(1-\tau)\right]^{\varepsilon}$ below the reference point and $R = n\left[w(1-\tau-\lambda)\right]^{\varepsilon}$ above it, where ε is the elasticity of the retirement age with respect to the implicit net-of-tax rate. Using the two tangency conditions of the marginal buncher, this implies that the excess mass at the reference point, $b = B/h_0(\hat{R})$, satisfies

$$\frac{b}{\hat{R}} = \left(\frac{1-\tau}{1-\tau-\lambda}\right)^{\varepsilon} - 1 \tag{6}$$

where $\lambda = \widetilde{\lambda}/w$ is the reference dependence parameter normalized by the wage per period.

4.3 Reference Point Formation

Prior literature has used the type of framework laid out above in order to estimate the strength of reference dependence around a given NRA. Our main goal is to study how individuals form reference points during a period in which the NRA is gradually changed by the government. Empirically, we observe bunching at the current NRA set by the government for each cohort, as well as residual bunching at the initial NRA of 65. This is informative about the process of reference point formation, suggesting that individuals adopt either the current or the initial NRA as their reference point. Alternative models of reference point formation considered in prior literature are less well in line with the dynamics we observe empirically. In particular, prior work has considered adaptive reference points as an average of past and current benchmarks (Baucells et al., 2011; DellaVigna et al., 2017; Thakral and Tô, 2021). In our setting, a similar process would entail reference points being a weighted average between past and current the NRAs, leading to a mass of excess retirements between the old and new NRA, but we instead find bifurcated bunching precisely at both thresholds. Moreover, we only we find stickiness of the initial NRA of 65, but no stickiness of any other NRA that was in place before an individual retires. This speaks against models of lagged expectations serving as reference points, as long as expectations update within a few years of the reform announcement (Kőszegi and Rabin, 2006, 2009; Smith, 2019; Heffetz, 2021).

With this discussion mind, we consider the following process of reference point formation:

$$\hat{R}_t = \begin{cases} NRA_t & \text{with probability } \pi_t \\ NRA_0 & \text{with probability } 1 - \pi_t \end{cases} \tag{7}$$

Thus, a fraction π_t of individuals in cohort t adopt the new NRA legislated by the government as their reference point, whereas the remaining fraction $1 - \pi_t$ still have the old NRA as their reference point.

We can now derive expressions for the excess mass at the old and new NRA. Denote by $\bar{b}_t$ the excess mass that would arise at the new NRA if all individuals in cohort t had fully adjusted their reference point. Bunching at the new NRA in cohort t is then given by

$$\frac{b_t^t}{NRA_t} = \pi_t \frac{\bar{b}_t}{NRA_t} \tag{8}$$

Under the additional assumption that λ and ε are locally independent of earnings ability n and independent of which reference point an individual adopts,[9] residual bunching at the old NRA is given by

$$\frac{b_0^t}{NRA_0} = (1 - \pi_t) \frac{\bar{b}_t}{NRA_t} \tag{9}$$

Combining equations (8) and (9) yields

$$\frac{\pi_t}{1 - \pi_t} = \frac{b_t^t / NRA_t}{b_0^t / NRA_0} \tag{10}$$

In words, equation (10) implies that the fraction of individuals who have adopted the new NRA as their reference point can be inferred from the relative magnitude of bunching responses at the new and the old NRA. Intuitively, since local retirement responses to the respective reference point are governed by the same individual-level parameters, observed bunching is informative about the relative number of individuals adopting each threshold as their reference point.

Equation (10) thus enables us to calibrate the fraction of individuals who have adopted the new NRA as their reference point in each cohort t, using the estimated excess masses at the new and old NRA as empirical moments. Importantly, this calibration does not require knowledge of the structural parameters λ and ε, but only the independence assumption stated above. We also note that our calibration allows for λ, ε, and τ to change across cohorts, since π_t is identified from relative bunching within the same cohort. This feature is particularly useful in our setting, as the 2006 reform changed financial retirement incentives for younger cohorts, which potentially affects the counterfactual retirement age distribution.

[9] These assumption are in line with the common assumptions in the literature. Prior work on reference dependence typically considers λ to be homogenous, and the bunching literature commonly assumes a locally constant ε. Note that we also require τ to be constant between the old and new NRA. This is guaranteed by our institutional setting, where there is no financial incentive kink around the NRA.

4.4 Calibration Results

Table 2 presents the calibration results. Columns (1) to (3) display the respective cohorts and NRA locations from which the reduced-form bunching estimates are obtained, Columns (4) and (5) show the excess masses at the old and new NRA estimated in Section 3.2, and Column (6) shows the fraction of individuals who have adopted the new NRA as their reference point, π_t, calibrated according to equation (10). In the first cohort group affected by the reform (t =1), facing an NRA of 65 and 1 month, we estimate a π_t of 0.64. In other words, 64% of individuals immediately adjust their reference to the new NRA, while 36% still have their reference point at the old NRA in the first year after the reform. The fraction of adjusted reference points then increases to 79% in the second post-reform cohort group and to 93% in the third cohort. In the fourth post-reform cohort group, the π_t reaches 98%, and it remains similar afterwards.

TABLE 2: CALIBRATION RESULTS

(1)	(2)	(3)	(4)	(5)	(6)
t	Birth Cohorts	NRA_t	b_0^t (Old NRA)	b_t^t (New NRA)	π_t
0	Jan. 1947 – Dec. 1947	65	22.55 (0.41)	–	–
1	Jan. 1948 – Nov. 1948	65 + 1 m.	8.69 (0.23)	15.66 (0.31)	0.642 (0.004)
2	Dec. 1948 – Oct. 1949	65 + 2 m.	5.57 (0.16)	20.61 (0.44)	0.785 (0.004)
3	Nov. 1949 – Sep. 1950	65 + 3 m.	2.25 (0.13)	32.02 (0.59)	0.933 (0.003)
4	Oct. 1950 – Jun. 1951	65 + 6 m.	0.65 (0.10)	35.75 (0.71)	0.982 (0.003)
5	Jul. 1951 – Mar. 1952	65 + 9 m.	0.82 (0.10)	33.86 (0.59)	0.975 (0.003)
6	Apr. 1952 – Dec. 1952	66	1.32 (0.11)	29.44 (0.54)	0.954 (0.004)
7	Jan. 1953 – Aug. 1955	66 + 4 m.	1.05 (0.06)	32.72 (0.36)	0.966 (0.002)

Notes: The table shows calibration results for the share of individuals who have adopted the new NRA as their reference point in each cohort group. Column (1) indicates the NRA cohort group and Column (2) the corresponding range of birth dates. Column (3) shows the NRA faced by each cohort group in years and months. Columns (4) and (5) show the estimated excess mass at the old NRA of 65 and at the new NRA, respectively (see Section 3.2). Column (6) shows the calibrated share of individuals with a reference point at the new NRA, π_t, obtained from equation (10). In Columns (4) to (6), bootstrapped standard errors are in parentheses.

These calibration results quantitatively corroborate our reduced-form findings. We find that individuals adopt the new NRA as a reference point relatively quickly. Even though the NRA had been set at 65 years for decades, around two thirds of workers adopt the newly legislated NRA of 65 years and 1 month already in the first post-reform year. A significant but decreasing share of individuals still use the old NRA as a reference point within three years of the reform, and the new NRA is virtually fully adopted as a reference point in the fourth year since the reform starts to take effect.

5 Alternative Mechanisms

In the previous section, we show that the adjustment dynamics we document are well explained by workers gradually adopting the new NRA as their reference point. Prior literature has considered several alternative mechanisms that could in principle generate retirement bunching at the NRA. In this section, we discuss these mechanisms in turn and we argue that none of them can account for the set of empirical patterns we document.

Financial Incentives. A first potential explanation for retirement bunching is that the NRA could coincide with a change in financial incentives. This is not the case in our setting. First-pillar pensions are paid automatically from the NRA, independently of whether an individual continues to work (see Section 2.1). Hence, retiring at the NRA does not change first-pillar pension wealth, and there is no earnings test that would penalize continued work beyond the NRA. Second-pillar pensions can also be claimed independently of the NRA throughout our analysis period. Moreover, for post-reform cohorts, the old NRA of 65 is not linked to any retirement or benefit rules at all. Financial incentives can thus neither explain bunching at the new NRA nor residual bunching at the old NRA.

Liquidity Constraints. A second potential mechanism that has been discussed in the literature is liquidity constraints. Since first-pillar pensions only become available at the NRA, workers with little liquid wealth and limited second-pillar entitlements may be unable to retire before the NRA, which could generate bunching at the NRA. While liquidity constraints could thus in principle contribute to bunching at the new NRA, existing empirical evidence suggests a limited role for this mechanism. For instance, Rabaté et al. (2024) find that bunching at the NRA in the Netherlands remains high among workers with high liquid wealth. Furthermore, liquidity constraints cannot explain residual bunching at the old NRA since first-pillar pension income does not become available at age 65 among post-reform cohorts.

Firm Responses. A third potential mechanism is given by firm responses. In the Netherlands, as in some other countries, many employment contracts terminate automatically at the NRA, and employers can dismiss workers at the NRA without the usual employment protection (see Section 2.1). Such firm-side factors could contribute to bunching at the NRA. However, prior evidence suggests that firm-side factors such as mandatory retirement explain only a limited part of retirement responses to the NRA (Rabaté, 2019; Seibold, 2021; Rabaté et al., 2024). Furthermore, firm responses cannot account for residual bunching at the old NRA, since the lifting of employment protection is tied to the new NRA rather than to age 65. Similarly, it is not clear why firm responses should lead to the increase of bunching at the new NRA that we observe over time.

Information and Learning. Finally, the gradual adjustment we document could in principle reflect incomplete information. If workers in the first post-reform cohorts were not aware of the

NRA increase, some of them may retire at 65 because they mistake this age for the NRA. However, several features of the Dutch setting make this explanation unlikely. First, as explained in Section 2.1, the NRA is highly salient in official communication, and information campaigns specifically addressed the NRA increase. For instance, an online tool where workers can readily find their individual NRA was launched as part of the national "Pension Days" campaign in 2013, the first year of the increase. Second, every individual is contacted by the social security administration in the year before reaching their NRA. Hence, even workers who were initially unaware of the reform would learn about their new NRA before reaching age 65. Third, the NRA increase was the subject of intense public debate, which makes it unlikely that older workers close to retirement were unaware of it. Finally, survey evidence suggests that Dutch workers are well-informed about the NRA and update their retirement plans in response to announced changes (e.g. Alessie et al., 2011; de Grip et al., 2013).[10]

6 Conclusion

In this paper, we study how retirement behavior adjusts to an increase in the Normal Retirement Age. We find that a Dutch reform that increased the NRA from 65 to 67 had a large impact on retirement behavior, driven by the retirement spike at the NRA moving in lockstep with the legislated NRA. However, adjustment is not immediate. In the short run, retirement behavior remains partly anchored at the old NRA: bunching at the new NRA builds up gradually, and a sizable spike persists at the old NRA of 65 in the first years of the reform. We interpret these dynamics through the lens of a model of reference-dependent retirement behavior, in which workers gradually adopt the new NRA as their reference point. A simple structural calibration implies that about a third of workers still hold the old NRA as their reference point in the short run, but that the new NRA is almost fully adopted as a reference point after three years.

Our findings have important implications for pension policy. Prior research has shown that the NRA has large effects on retirement behavior, but how quickly individuals respond to reforms that change the NRA has been less clear. We show that responses to an NRA increase are indeed somewhat muted in the short run, but retirement behavior adjusts fully to the new NRA after only a few years. This matters for the dynamics of the fiscal effects of pension reforms, which policymakers seeking to stabilize public pension finances need to anticipate.

Our results also have implications for the literature on reference-dependent preferences. We show that public policy can effectively influence reference points for retirement. During reform periods in which both the old and the new NRA provide salient potential benchmarks for the retirement age, individuals adopt the new policy as their reference point relatively quickly. Moreover, our findings suggest that reference points adapt in a discrete way with individuals adopting

[10] Evidence from other settings likewise suggests that older workers are well-informed about their NRA. For instance, Caplin et al. (2022) find that, while younger workers in Denmark are uncertain about the future NRA, individuals close to retirement have accurate beliefs about their NRA even during periods of ongoing pension reforms.

either the old or the new NRA. These findings speak to how reference points are formed and updated, which has been a central contested question in prior literature.

References

ABELER, J., A. FALK, L. GOETTE, AND D. HUFFMAN (2011): "Reference Points and Effort Provision," *American Economic Review*, 101, 470–492.

ALESSIE, R., M. VAN ROOIJ, AND A. LUSARDI (2011): "Financial Literacy, Retirement Preparation and Pension Expectations in the Netherlands," *NBER Working Paper*.

ALLEN, E. J., P. M. DECHOW, D. G. POPE, AND G. WU (2017): "Reference-Dependent Preferences: Evidence from Marathon Runners," *Management Science*, 63, 1657–1672.

ANDERSEN, S., C. BADARINZA, L. LIU, J. MARX, AND T. RAMADORAI (2022): "Reference Dependence in the Housing Market," *American Economic Review*, 112, 3398–3440.

BARTLING, B., L. BRANDES, AND D. SCHUNK (2015): "Expectations as Reference Points: Field Evidence from Professional Soccer," *Management Science*, 61, 2646–2661.

BAUCELLS, M., M. WEBER, AND F. WELFENS (2011): "Reference-Point Formation and Updating," *Management Science*, 57, 506–519.

BEHAGHEL, L. AND D. M. BLAU (2012): "Framing Social Security Reform: Behavioral Responses to Changes in the Full Retirement Age," *American Economic Journal: Economic Policy*, 4, 41–67.

BROWN, K. M. (2013): "The Link between Pensions and Retirement Timing: Lessons from California Teachers," *Journal of Public Economics*, 98, 1–14.

BURTLESS, G. (1986): "Social Security, Unanticipated Benefit Increases, and the Timing of Retirement," *Review of Economic Studies*, 53, 781–805.

CAMERER, C., L. BABCOCK, G. LOEWENSTEIN, AND R. THALER (1997): "Labor Supply of New York City Cabdrivers: One Day at a Time," *Quarterly Journal of Economics*, 112, 407–441.

CAPLIN, A., E. LEE, S. LETH-PETERSEN, AND J. SÆVERUD (2022): "Communicating Social Security Reform," NBER Working Paper 30645, National Bureau of Economic Research.

CHETTY, R., J. N. FRIEDMAN, T. OLSEN, AND L. PISTAFERRI (2011): "Adjustment Costs, Firm Responses, and Micro vs. Macro Labor Supply Elasticities: Evidence from Danish Tax Records," *Quarterly Journal of Economics*, 126, 749–804.

CRAWFORD, V. P. AND J. MENG (2011): "New York City Cab Drivers' Labor Supply Revisited: Reference-Dependent Preferences with Rational-Expectations Targets for Hours and Income," *American Economic Review*, 101, 1912–1932.

CRIBB, J., C. EMMERSON, AND G. TETLOW (2016): "Signals Matter? Large Retirement Responses to Limited Financial Incentives," *Labour Economics*, 42, 203–212.

DE GRIP, A., D. FOUARGE, AND R. MONTIZAAN (2013): "How Sensitive are Individual Retirement Expectations to Raising the Retirement Age?" *De Economist*, 161, 225–251.

DELLAVIGNA, S., A. LINDNER, B. REIZER, AND J. F. SCHMIEDER (2017): "Reference-Dependent Job Search: Evidence from Hungary," *Quarterly Journal of Economics*, 132, 1969–2018.

DESHPANDE, M., I. FADLON, AND C. GRAY (2024): "How Sticky Is Retirement Behavior in the United States? Responses to Changes in the Full Retirement Age," *American Economic Review: Insights*, 6, 370–387.

DOLLS, M. AND C. KROLAGE (2023): "'Earned, not given'? The Effect of Lowering the Full Retirement Age on Retirement Decisions," *Journal of Public Economics*, 223, 104909.

DUGGAN, M., I. DUSHI, S. JEONG, AND G. LI (2023): "The Effects of Changes in Social Security's Delayed Retirement Credit: Evidence from Administrative Data," *Journal of Public Economics*, 223, 104899.

FARBER, H. S. (2008): "Reference-Dependent Preferences and Labor Supply: The Case of New York City Taxi Drivers," *American Economic Review*, 98, 1069–1082.

GELBER, A. M., D. JONES, AND D. W. SACKS (2020): "Estimating Adjustment Frictions Using Nonlinear Budget Sets: Method and Evidence from the Earnings Test," *American Economic Journal: Applied Economics*, 12, 1–31.

GENESOVE, D. AND C. MAYER (2001): "Loss Aversion and Seller Behavior: Evidence from the Housing Market," *Quarterly Journal of Economics*, 116, 1233–1260.

GILL, D. AND V. PROWSE (2012): "A Structural Analysis of Disappointment Aversion in a Real Effort Competition," *American Economic Review*, 102, 469–503.

GIUPPONI, G. AND A. SEIBOLD, eds. (2024): *Rethinking Pension Reform*, Paris & London: CEPR Press.

GRUBER, J., O. KANNINEN, AND T. RAVASKA (2022): "Relabeling, Retirement and Regret," *Journal of Public Economics*, 211, 104677.

HEFFETZ, O. (2021): "Are Reference Points Merely Lagged Beliefs Over Probabilities?" *Journal of Economic Behavior & Organization*, 181, 252–269.

HOMONOFF, T. A. (2018): "Can Small Incentives Have Large Effects? The Impact of Taxes Versus Bonuses on Disposable Bag Use," *American Economic Journal: Economic Policy*, 10, 177–210.

KLEVEN, H. J. (2016): "Bunching," *Annual Review of Economics*, 8, 435–464.

KLEVEN, H. J. AND M. WASEEM (2013): "Using Notches to Uncover Optimization Frictions and Structural Elasticities: Theory and Evidence from Pakistan," *Quarterly Journal of Economics*, 128, 669–723.

KNOEF, M., J. BEEN, R. ALESSIE, K. CAMINADA, K. GOUDSWAARD, AND A. KALWIJ (2016): "Measuring Retirement Savings Adequacy: Developing a Multi-Pillar Approach in the Netherlands," *Journal of Pension Economics & Finance*, 15, 55–89.

KŐSZEGI, B. AND M. RABIN (2006): "A Model of Reference-Dependent Preferences," *Quarterly Journal of Economics*, 121, 1133–1165.

——— (2007): "Reference-Dependent Risk Attitudes," *American Economic Review*, 97, 1047–1073.

——— (2009): "Reference-Dependent Consumption Plans," *American Economic Review*, 99, 909–936.

LALIVE, R., A. MAGESAN, AND S. STAUBLI (2023): "How Social Security Reform Affects Retirement and Pension Claiming," *American Economic Journal: Economic Policy*, 15, 115–150.

LINDEBOOM, M. AND R. MONTIZAAN (2020): "Disentangling Retirement and Savings Responses," *Journal of Public Economics*, 192, 104297.

LUMSDAINE, R. L., J. H. STOCK, AND D. A. WISE (1996): "Why Are Retirement Rates So High at Age 65?" in *Advances in the Economics of Aging*, ed. by D. A. Wise, University of Chicago Press, 61–82.

MANOLI, D. AND A. WEBER (2016): "Nonparametric Evidence on the Effects of Financial Incentives on Retirement Decisions," *American Economic Journal: Economic Policy*, 8, 160–182.

MASTROBUONI, G. (2009): "Labor Supply Effects of the Recent Social Security Benefit Cuts: Empirical Estimates Using Cohort Discontinuities," *Journal of Public Economics*, 93, 1224–1233.

MAVROKONSTANTIS, P. AND A. SEIBOLD (2022): "Bunching and Adjustment Costs: Evidence from Cypriot Tax Reforms," *Journal of Public Economics*, 214, 104727.

O'DONOGHUE, T. AND C. SPRENGER (2018): "Reference-Dependent Preferences," in *Handbook of Behavioral Economics: Applications and Foundations 1*, ed. by B. D. Bernheim, S. DellaVigna, and D. Laibson, North-Holland, vol. 1, 1–77.

OECD (2025): *Pensions at a Glance 2025: OECD and G20 Indicators*, Paris: OECD Publishing.

ORAL, E., S. RABATÉ, AND A. SEIBOLD (2026): "The Social Multiplier of Pension Reform," *CEPR Discussion Paper*.

POPE, D. G. AND M. E. SCHWEITZER (2011): "Is Tiger Woods Loss Averse? Persistent Bias in the Face of Experience, Competition, and High Stakes," *American Economic Review*, 101, 129–157.

RABATÉ, S. (2019): "Can I Stay or Should I Go? Mandatory Retirement and the Labor-Force Participation of Older Workers," *Journal of Public Economics*, 180, 104078.

RABATÉ, S., E. JONGEN, AND T. ATAV (2024): "Increasing the Retirement Age: Policy Effects and Underlying Mechanisms," *American Economic Journal: Economic Policy*, 16, 259–291.

RECK, D. AND A. SEIBOLD (2026): "The Welfare Economics of Reference Dependence," *American Economic Journal: Applied Economics*, 18, 178–219.

REES-JONES, A. (2018): "Quantifying Loss-Averse Tax Manipulation," *Review of Economic Studies*, 85, 1251–1278.

SAEZ, E. (2010): "Do Taxpayers Bunch at Kink Points?" *American Economic Journal: Economic Policy*, 2, 180–212.

SEIBOLD, A. (2021): "Reference Points for Retirement Behavior: Evidence from German Pension Discontinuities," *American Economic Review*, 111, 1126–1165.

——— (2024): "How to Effectively Encourage Later Retirement? Statutory Retirement Ages versus Financial Incentives," in *Rethinking Pension Reform*, ed. by G. Giupponi and A. Seibold, Paris & London: CEPR Press, 25–32.

SMITH, A. (2019): "Lagged Beliefs and Reference-Dependent Utility," *Journal of Economic Behavior & Organization*, 167, 331–340.

STAUBLI, S. AND J. ZWEIMÜLLER (2013): "Does Raising the Early Retirement Age Increase Employment of Older Workers?" *Journal of Public Economics*, 108, 17–32.

THAKRAL, N. AND L. T. TÔ (2021): "Daily Labor Supply and Adaptive Reference Points," *American Economic Review*, 111, 2417–2443.

UNITED NATIONS (2024): "World Population Prospects 2024," United Nations, Department of Economic and Social Affairs, Population Division.

VERMEER, N., M. VAN ROOIJ, AND D. VAN VUUREN (2019): "Retirement Age Preferences: The Role of Social Interactions and Anchoring at the Statutory Retirement Age," *De Economist*, 167, 307–345.

ZARESANI, A. (2020): "Adjustment Costs and Incentives to Work: Evidence from a Disability Insurance Program," *Journal of Public Economics*, 188, 104223.

A Appendix Figures and Tables

TABLE A1: SUMMARY STATISTICS BY COHORT GROUP

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Cohort group	$t = 0$	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$	$t = 6$	$t = 7$
	(Jan 1942– Dec 1947)	(Jan 1948– Nov 1948)	(Dec 1948– Oct 1949)	(Nov 1949– Sep 1950)	(Oct 1950– Jun 1951)	(Jul 1951– Mar 1952)	(Apr 1952– Dec 1952)	(Jan 1953– Dec 1954)
NRA	65	65+1m	65+2m	65+3m	65+6m	65+9m	66	66+4m
Female	0.34	0.37	0.38	0.39	0.39	0.40	0.41	0.42
	(0.47)	(0.48)	(0.48)	(0.49)	(0.49)	(0.49)	(0.49)	(0.49)
Married/civil partnership	0.80	0.78	0.76	0.76	0.75	0.74	0.74	0.73
	(0.40)	(0.42)	(0.43)	(0.43)	(0.43)	(0.44)	(0.44)	(0.44)
Dutch-born	0.88	0.86	0.86	0.85	0.85	0.85	0.85	0.85
	(0.33)	(0.35)	(0.35)	(0.35)	(0.36)	(0.36)	(0.35)	(0.36)
Labor income at 57 (EUR)	30,672	32,374	33,666	35,010	35,885	36,338	37,055	37,822
	(34,008)	(34,658)	(45,113)	(63,801)	(38,999)	(35,231)	(42,854)	(44,691)
Year of birth	1945.26	1948.41	1949.33	1950.25	1951.09	1951.84	1952.58	1953.96
	(1.72)	(0.26)	(0.26)	(0.26)	(0.21)	(0.22)	(0.21)	(0.58)
Retirement age	62.81	63.29	63.48	63.85	64.02	64.14	64.28	64.54
	(3.32)	(3.14)	(3.04)	(2.97)	(2.99)	(3.03)	(3.06)	(3.10)
Retirement at the NRA	0.10	0.15	0.16	0.22	0.23	0.22	0.22	0.24
	(0.31)	(0.35)	(0.37)	(0.41)	(0.42)	(0.42)	(0.42)	(0.43)
Retirement before the NRA	0.65	0.61	0.60	0.54	0.53	0.53	0.53	0.52
	(0.48)	(0.49)	(0.49)	(0.50)	(0.50)	(0.50)	(0.50)	(0.50)
Retirement after the NRA	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24
	(0.43)	(0.43)	(0.43)	(0.43)	(0.43)	(0.43)	(0.43)	(0.43)
Observations	713,890	134,063	133,376	133,769	110,856	114,696	117,532	320,595

Notes: The table shows summary statistics separately for each cohort group. Groups are defined by the NRA that applies to the respective birth cohorts, as indicated in the table header. Each cell shows the mean, with the standard deviation in parentheses. Retirement age is the age of last employment exit. Income figures are rounded to full euros.